MedViz: Illuminating Biomedical Literature using Agentic AI and Visual Analytics

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Introduction

In the rapidly evolving field of biomedicine, researchers are inundated with an ever-growing corpus of publications. This massive biomedical knowledge poses a significant challenge in terms of efficient data navigation and exploration [1]. Current biomedical literature search engines primarily present a list of relevant articles for a given query, and usually don't provide a global landscape of the knowledge space of interest. Users can browse the returned articles but typically have limited functionality for effective navigation and in-depth exploration, making it challenging to answer complex research questions. While recent generative AI technologies, such as LLMs, have shown considerable promise in assisting with tasks like extraction and summarization, their traditional implementations usually provide only dialog-like responses. These responses are not inherently integrated into interactive workflows, resulting in a loss of contextual continuity and necessitating manual processes, such as copy-pasting between different systems. This limitation significantly reduces researchers' efficiency in gaining cohesive insights within a unified environment.

To address these challenges, we developed a new tool called MedViz, which leverages LLM-based agents and visual analytics technologies to provide a new way for biomedical researchers to explore the semantic space of biomedical literature, e.g., identify topic clusters and track temporal research trends. Moreover, MedViz integrates advanced hybrid agents combining generative AI and visual analytics capabilities, enabling users to interact through both natural language queries and visual actions seamlessly. These agents not only assist in general tasks such as literature search and information extraction, but also complex tasks such as automatic summarization and semantic space discovery, enabling deeper exploration of relationships and connections within biomedical data. Unlike conventional chatbot-based systems, MedViz's hybrid agents embed their responses into the interactive workflow, maintaining contextual continuity and enhancing the interpretability of the exploratory process.

Methods

As shown in Figure 1, MedViz consists of three main components, data pipeline, LLM-based agents, and user interface.

The ***data pipeline*** aggregates the data related to biomedical literature from PubMed, iCite, and PubTator based on user-specified PubMed query to generate a dataset for further analysis. After collecting and pre-processing the raw metadata, titles and abstracts are embedded into high-dimensional vectors using an LLM embedder. These vectors are reduced to 2D or 3D coordinates for visualization. The coordinates are then clustered, and LLMs automatically generate topic labels for each cluster to facilitate exploration.

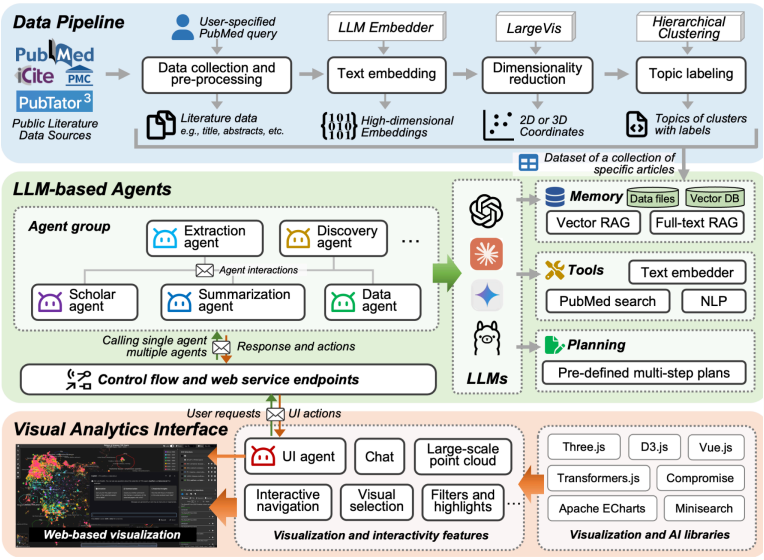

Figure 1. Overall architecture of MedViz, which consists of three major components: Data Pipeline, LLM-based Agents, and Visual Analytics Interface.

The **LLM-based agents** component integrates multiple specialized agents for specific tasks, such as the Data agent for fetching extra references and the Extraction agent for information retrieval. Specifically, the Scholar agent utilizes retrieval-augmented generation (RAG) to produce citation-backed synthesized responses, while the Discovery agent leverages embeddings and customized LLMs to generate potential research hypotheses directly from the semantic space. These agents can operate independently or collaboratively, guided by a control flow module that manages requests and responses. Built upon LLMs, agents integrate external modules to enhance their capabilities, including: 1) memory module providing

external information based on RAG; 2) tools module for executing specific tasks; and 3) planning module that provides prompts of predefined multi-step plans to improve efficiency and consistency in complex tasks.

Beyond the LLM-based agents, a UI agent was integrated into the visual analytics interface. This agent facilitates communication with other LLM-based agents, leveraging user requests and contextual information such as article selections, semantic areas of focus, and interaction history. This allows for more precise and contextually relevant responses. In turn, the UI agent also receives instructions from backend agents, enabling it to execute specific visual interactions alongside text responses. For instance, the UI agent can perform actions like zooming, filtering, and labeling on the semantic map to visually represent the outputs of complex backend agents. Ultimately, the UI agent's goal is to improve interpretability by offering users immediate and contextual visual feedback.

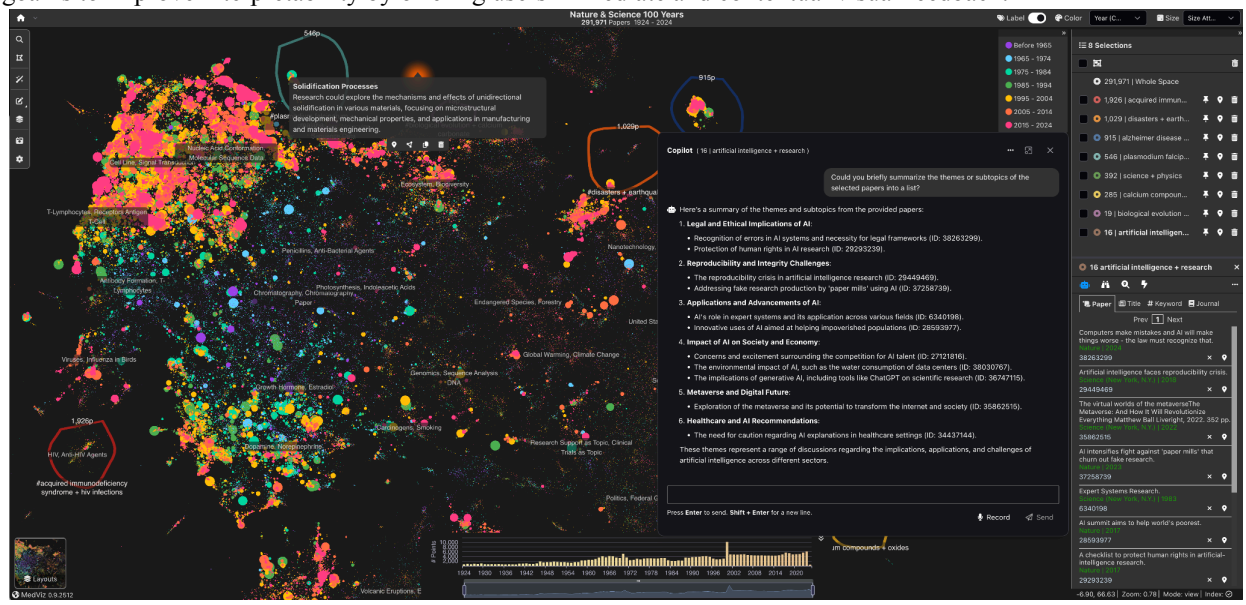

Figure 2. Screenshot of MedViz, showing the semantic map with topic labels and user-created selections.

The **visual analytics interface** employs WebGL to visualize the dataset generated by the data pipeline as a semantic point cloud map, where each point corresponds to an individual article. Each point's color encodes the publication year, while the point size encodes the citation number, indicating the impact of each article. The spatial clustering of points indicates semantic similarities and suggests underlying topical groupings. Additionally, MedViz displays topic labels for each identified cluster at multiple levels of detail, aiding users in quickly understanding the themes present in the dataset. Users can interact extensively with this semantic map to explore areas of interest more closely. For example, users can visually select articles using polygonal selections; execute keyword-based searches and highlight the filtered results; adjust the time range to observe temporal trends and shifts in research focus; and click on the topic labels to isolate and examine specific collections of articles.

To obtain deeper insights from the semantic map, users can interact with specialized agents via a chat interface or pointer tools. As shown in Figure 2, user queries, along with relevant contextual information (e.g., selected articles) collected by the UI agent, are forwarded to backend agents by the control flow based on predefined plans. For instance, the Scholar agent uses RAG to generate citation-backed responses, while the Discovery agent employs semantic embeddings to suggest research hypotheses. Both agents may call the Data agent to retrieve more information if needed. Responses from these backend agents are relayed back to the UI agent, which then visually presents these insights on the interface.

Discussion and Future Work

MedViz, including sample datasets, can be accessed at https://medviz.org/. We validated our system through case studies with domain experts, who performed tasks such as research topic analysis and information extraction. Expert feedback confirmed that MedViz effectively uncovered novel insights, significantly assisting in topic identification and literature synthesis. Experts also expressed interest in utilizing MedViz for more specialized domain-specific tasks and exploring larger datasets. Our future plans involve enhancing MedViz's scalability and functionality by integrating additional specialized agents for tasks like knowledge graph inference and multi-modal data analysis.

Reference

1. Bornmann, L., Haunschild, R. & Mutz, R. Growth rates of modern science: a latent piecewise growth curve approach to model publication numbers from established and new literature databases. Humanit Soc Sci Commun 8, 224 (2021).